

the vast majority of its prompts do not allow for multiple valid answers. During processing, we first used Llama Guard 3 (Chi et al., 2024) to filter out inappropriate user prompts. A small number of IPs contributed a disproportionately high number of interactions, leading to topic bias (e.g., towards anime); we deduplicated the dataset based on user IP to achieve a more even topic distribution. After preprocessing, we used GPT-4o as a classifier to select prompts expected to elicit diverse generations.³ To make sure that the selected prompts match the desired criteria, we manually labeled 100 prompts from the dataset (with 50 accepted and 50 rejected by GPT-4o), and found an 85% agreement between the classifier and the human labels. Randomly sampled prompts from this dataset are provided in Appendix A.3.

3.2 Evaluating Novelty

Models with diversity alone would not be sufficient for users: each of the individual generations should also be of high quality, with the ultimate goal of maximizing the cumulative utility of a user who interacts with the model. Thus, our evaluation framework aims to assess the extent to which the set of responses is mutually diverse and simultaneously high-quality, and our key metric directly models the cumulative utility for a user who observes a sequence of generations from a language model.

Partitioning the output space. Defining diversity with automatic metrics is challenging due to the complex nature of language. Traditional metrics like n-gram overlap or embedding similarity often fail to capture meaningful diversity, as they may either overemphasize trivial differences (e.g., paraphrasing) or miss important semantic distinctions. Prior work has attempted to measure diversity using lexical variation (Li et al., 2016), embedding distances (Zhang et al., 2019), or entropy-based measures (Dušek et al., 2020), but these approaches typically struggle to distinguish between functional diversity and surface-level variations that provide little additional value to users (Tevet & Berant, 2021).

Accordingly, to understand the diversity of the output space of language models, we propose a method that learns to partition the output space into equivalence classes from human annotations. Each class represents one unique generation that is roughly equivalent to the others in the same class and different from the generations in other classes.

Partitioning the output space boils down to defining a notion of *equivalence* between pairs of generations. Language model outputs can be trivially different from each other (e.g., in terms of style, tone, or length). We argue that true diversity should go beyond these surface-level differences, since such variations provide little utility to the user. Therefore, we mainly consider *functional equivalence* in the context of this work, which defines two generations to be different if and only if a user who has seen one generation would likely benefit from seeing the other. For example, in a story generation task, if two generations are retelling the original story with identical plot, but different character names, then we consider them to be functionally equivalent.

While this definition of functional equivalence is intuitive for human evaluators, it is non-trivial to capture this notion of equivalence using existing similarity metrics.⁴ To this end, the authors annotated 1,100 pairs of generations conditioned on prompts from both the NB-CURATED and NB-WILDCHAT datasets sampled from a diverse set of models (see Table 1 for a list of models). For annotation, we developed guidelines that focused on identifying whether two generations would provide distinct value to a user, rather than just surface-level differences.

From these annotated pairs, we used 1,000 for training and fine-tuned a deberta-v3-large model (He et al., 2021) to predict binary functional equivalence between two generations. To ensure that the classifier matches human judgment, we evaluated it on a held-out test set

³Prompt selection criteria are provided in Appendix A.2.

⁴We experimented with existing metrics such as BLEU (Papineni et al., 2002), ROUGE (Lin, 2004) and BERTScore (Zhang et al., 2019), but found their predictions to diverge from human judgment. See results in Appendix A.4.

of 100 pairs. The classifier achieved 79% accuracy compared to human labels, with an F1 score of 0.811, indicating strong performance on both positive and negative classes.

With the equivalence classifier, we can partition the output space into equivalence classes. For each new generation, we compare it against a random generation from each existing class. If the classifier determines it is functionally equivalent to any existing class, we assign it to the first such class found; otherwise, we create a new class with this generation. We define the distinct_k metric as the number of equivalence classes in a partition of k sample generations from a language model:

$$\text{distinct}_k := |\{c_i \mid i \in [k]\}| \quad (1)$$

where c_i is the equivalence class that the i -th generation belongs to. For a sufficiently large k , distinct_k quantifies the number of meaningfully unique generations that a user could have observed by sampling from the model k times. A higher score indicates that the model is better at producing meaningful alternatives to the most likely generation.

Unified measure of diversity and quality. How do we measure the cumulative utility of a sequence of generations $s_1, s_2, \dots, s_k$? To achieve this, we first need a model of user behavior that describes how users interact with and consume language model generations. Inspired by the user model underlying the rank-biased precision metric in information retrieval (Moffat & Zobel, 2008), we assume that the user has a patience level $p \in [0, 1]$: after observing each additional generation, they have a probability p of requesting an additional generation from the language model and observing the next generation, and a probability $1 - p$ of stopping interacting with the model. In other words, each additional generation should have a geometrically decaying effect on utility since it is more likely that the user simply would not have observed it.

To model the marginal utility of a distinct generation, we assume that the utility of a generation that is functionally equivalent to a previous generation is zero. Under our user model with patience p , the cumulative utility of a sequence of generations is given by:

$$\text{utility}_k := \frac{1-p}{1-p^k} \sum_{i=1}^k p^{i-1} \cdot \underbrace{\mathbb{1}[c_i \neq c_j, \forall j < i]}_{\text{marginal utility of generation } i} \cdot u_i \quad (2)$$

where c_i is the equivalence class of the i -th generation, u_i is the utility of the i -th generation given by some utility function, and $\frac{1-p}{1-p^k}$ is a normalization factor. In our evaluation setup, we set the patience level to 0.8. Note that as $p \rightarrow 0$, the metric converges to quality evaluation of a single generation, the de facto standard in language model evaluation.

Computing utility. We use the Skywork-Reward-Gemma-2-27B-v0.2 model (Liu et al., 2024) to score the quality of individual generations. We chose this reward model because it is one of the top performing models on RewardBench (Lambert et al., 2024), and outperforms larger models such as Nvidia’s 70B Nemotron model (Wang et al., 2024).

The reward scores produced by the reward model are real-valued and difficult to interpret for humans. To map these scores into a more interpretable utility measure that outputs values in $\{1, 2, \dots, 10\}$, we computed the distribution of reward values for 2,400 model generations on MT-Bench (Zheng et al., 2023) and used quality scores judged by GPT-4 to calibrate the reward mapping to match the score distributions.

4 Evaluation Results on NOVELTYBENCH

4.1 Models considered

We are particularly interested in the behavior of “aligned” language models which have been adapted using supervised finetuning and/or reinforcement learning to be conversational, instruction-following, and helpful. In particular, we consider a list of 20 state-of-the-art

Model Provider	Variants	Distinct	Utility
Anthropic (Anthropic, 2024)	Claude-3.5 Haiku	1.94	2.50
	Claude-3.5 Sonnet	1.76	2.36
	Claude-3 Opus	2.04	2.67
OpenAI (OpenAI, 2024)	gpt-4o-mini	2.65	3.11
	gpt-4o	2.88	3.27
Gemini (Google, 2024)	gemini-1.5-pro	1.85	2.73
	gemini-2.0-flash-lite	2.83	3.20
	gemini-2.0-flash	2.81	3.17
	gemini-2.0-pro	2.25	2.64
Cohere (Cohere, 2024)	command-r7b	3.58	3.35
	command-r	2.68	2.98
	command-r-plus	2.79	3.08
Gemma 2 (Gemma Team et al., 2024)	gemma-2-2b-it	5.66	4.63
	gemma-2-9b-it	3.25	3.93
	gemma-2-27b-it	3.03	3.77
Llama 3 (Llama Team et al., 2024)	Llama-3.2-1B	6.74	2.81
	Llama-3.2-3B	5.10	3.24
	Llama-3.1-8B	5.24	3.76
	Llama-3.3-70B	2.49	2.87
	Llama-3.1-405B	3.20	3.39

Table 1: List of models used as baselines for our benchmark, with their average number of distinct generations and cumulative utility scores out of 10 generations.

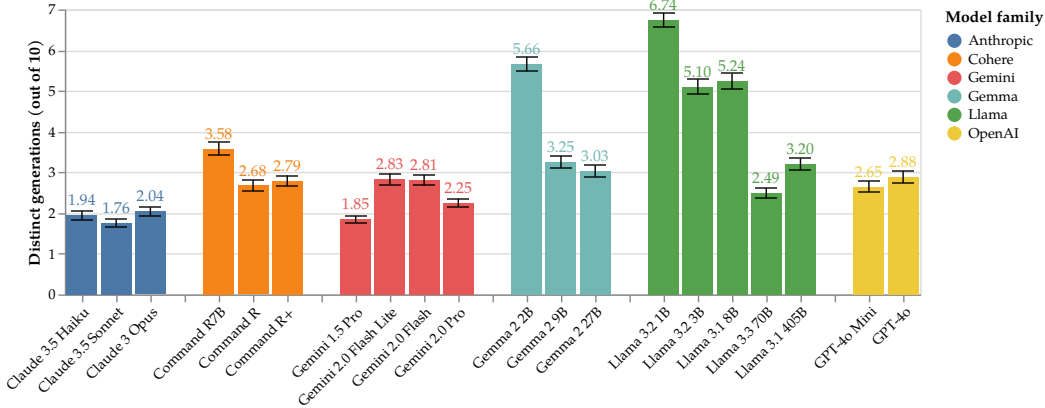


Figure 2: Average number of unique generations out of a sample of 10 for all prompts in NOVELTYBENCH. Error bars indicate 95% confidence intervals.

models from Anthropic, OpenAI, Google and Meta (listed in Table 1) as a representative sample of the performance of frontier models.

Where applicable, we disable retrieval-augmented generation and tool-use. While these augmentations undoubtedly influence language model diversity, examining their effects falls outside our scope. Instead, we focus our ablations on how various prompting techniques affect output diversity (Section 4.3). We evaluate model output distributions by independently sampling 10 generations with a temperature of 1—representing the best-case scenario for diversity, as most APIs default to lower temperatures or nucleus sampling.

4.2 State-of-the-art models struggle to generate diverse and useful outputs

Our evaluation reveals that frontier language models struggle to produce simultaneously diverse and useful responses. As shown in Figure 2, models like Claude 3 and GPT-4o